TASK 3 – Choose Data Types and Check Type Compatibility

1. Data Type Selection

System value	Selected type	Reason
Fare in paise	long long	Stores exact money values and supports large totals
Smart-card balance	long long	Prevents overflow during balance calculations
Peak-hour status	bool	Stores true or false
Zone number	int	Stores small numeric zone identifiers
Zone/station code	char	Stores a single character
Hour of travel	int	Stores hours from 0 to 23
Daily journey count	long long	Supports future network growth
Percentage/rate calculations	double	Supports decimal calculations
Approximate measurement	float	Uses less storage where high precision is unnecessary
Automatically deduced values	auto	Allows the compiler to determine the type

The actual sizes are system-dependent and are displayed by the implementation using sizeof.

2. Why Integer Paise Is Better Than Double Rupees

Money should be stored as integer paise rather than double rupees because binary floating-point cannot represent every decimal value exactly.

For example:

- Rs. 10.50 = 1050 paise
- Rs. 20.00 = 2000 paise
- Rs. 30.00 = 3000 paise

Integer paise provides exact addition and subtraction without floating-point rounding errors. Therefore, long long is selected for fares, balances and accumulated collections.

3. Type Compatibility

C++ performs automatic conversions when different data types are used in an expression.

- int combined with double: the integer is converted to double.
- int combined with bool: zero becomes false; a non-zero value becomes true.
- Integer division discards the fractional part.
- char can be promoted to int during arithmetic.
- Mixing signed and unsigned integers can convert the signed value to unsigned, producing unexpected comparisons.

Explicit conversion using static_cast makes the programmer's intention clearer.

4. Integer Limits and Network Collection

The <climits> header provides limits such as INT_MIN, INT_MAX, LLONG_MIN and LLONG_MAX.

At 3,000,000 journeys per day, assuming a maximum base fare of Rs. 30:

Total collection = 3,000,000 × 3,000 paise = 9,000,000,000 paise.

This exceeds the usual 32-bit signed int maximum of 2,147,483,647. Therefore, int is not sufficient for this paise total. long long is a safer choice.

5. Alternative: Double Rupees

If money were stored using double, each addition would still take O(1) time. Both double and long long commonly occupy 8 bytes on the target system, which the program verifies using sizeof.

However, repeated floating-point additions can accumulate rounding differences. Integer paise avoids this problem for monetary values.

Comparison	**Integer paise (long long)**	**Double rupees**
Accuracy	Exact for whole paise	Possible rounding error
Addition time	O(1)	O(1)
Typical storage	8 bytes	8 bytes
Suitability	Recommended for money	Not recommended for exact fare accounting

Conclusion: The metro system should use long long for money and large totals, int for hours and zone numbers, and bool for conditions. This design provides accurate fare calculations and supports network growth.